

COM661 — Back-End API Development

Postman API Documentation

Collection	COM-661
Base URL	http://127.0.0.1:5001
Authentication	Basic Auth (login) + JWT Bearer Token {{token}}
Total Requests	26
Published Docs	https://documenter.getpostman.com/view/47627997/2sBXiesZVT
Framework	Flask (Python) + MongoDB

1. Authentication

1.1 Login User

Method	URL	Auth
POST	http://127.0.0.1:5001/users/login	Basic Auth

AUTH: Username = test@email.com | Password = password123

POST-RESPONSE SCRIPT:

```
var response = pm.response.json();
pm.collectionVariables.set("token", response.token);
pm.collectionVariables.set("refresh_token", response.refresh_token);
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Response has token", function () {
  pm.expect(response).to.have.property("token");
});
pm.test("Response has refresh token", function () {
  pm.expect(response).to.have.property("refresh_token");
});
pm.test("Role is admin", function () {
  pm.expect(response.role).to.eql("admin");
});
```

1.2 Register User

Method	URL	Auth
POST	http://127.0.0.1:5001/users/register	No Auth

REQUEST BODY:

```
{
  "name": "Test User",
  "email": "test@email.com",
  "password": "password123",
  "role": "user"
}
```

TEST SCRIPT:

```
pm.test("Status code is 400 (email already exists)", function () {
  pm.response.to.have.status(400);
});
```

2. Activity CRUD

2.1 Get All Activities

Method	URL	Auth
GET	http://127.0.0.1:5001/activities	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Returns array of activities", function () {
  var response = pm.response.json();
  pm.expect(response.data).to.be.an("array");
});
pm.test("Total is greater than 0", function () {
  var response = pm.response.json();
  pm.expect(response.total).to.be.above(0);
});
```

2.2 Get One Activity

Method	URL	Auth
GET	http://127.0.0.1:5001/activities/ACT-001	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Activity has correct ID", function () {
  var response = pm.response.json();
  pm.expect(response.data.activity_type).to.be.a("string");
});
```

2.3 Create Activity

Method	URL	Auth
POST	http://127.0.0.1:5001/activities	Bearer {{token}} (Inherit)

REQUEST BODY:

```
{
  "activity_type": "Transport",
  "activity_detail": "Car to gym",
  "activity_value": 10,
  "unit": "km",
  "emission_factor": 0.21,
  "timestamp": "2026-03-06"
}
```

TEST SCRIPT:

```
var response = pm.response.json();
if (response.data) {
  pm.collectionVariables.set("new_activity_id", response.data._id);
}
pm.test("Status code is 201", function () {
  pm.response.to.have.status(201);
});
pm.test("Activity created successfully", function () {
  pm.expect(response.message).to.eql("Activity logged successfully");
});
pm.test("Carbon emission auto calculated", function () {
  pm.expect(response.data).to.have.property("carbon_emission");
});
```

```
});
```

2.4 Update Activity

Method	URL	Auth
PUT	http://127.0.0.1:5001/activities/ACT-001	Bearer {{token}} (Inherit)

REQUEST BODY:

```
{
  "activity_detail": "Updated car travel",
  "activity_value": 150,
  "emission_factor": 0.21
}
```

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Activity updated successfully", function () {
  var response = pm.response.json();
  pm.expect(response.message).to.eql("Activity updated successfully");
});
```

2.5 Delete Activity

Method	URL	Auth
DELETE	http://127.0.0.1:5001/activities/{{delete_activity_id}}	Bearer {{token}} (Inherit)

PRE-REQUEST SCRIPT (creates temp activity so delete always works):

```
pm.sendRequest({
  url: 'http://127.0.0.1:5001/activities',
  method: 'POST',
  header: {
    'Content-Type': 'application/json',
    'Authorization': 'Bearer ' + pm.collectionVariables.get('token')
  },
  body: {
    mode: 'raw',
    raw: JSON.stringify({
      "activity_type": "Transport",
      "activity_detail": "Temp delete test",
      "activity_value": 10,
      "unit": "km",
      "emission_factor": 0.21
    })
  },
  function(err, res) {
    pm.collectionVariables.set("delete_activity_id", res.json()._id);
  });
```

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Activity deleted successfully", function () {
  var response = pm.response.json();
  pm.expect(response.message).to.eql("Activity deleted successfully");
});
```

3. Sources Sub-Documents

3.1 Add Source

Method	URL	Auth
POST	http://127.0.0.1:5001/activities/ACT-001/sources	Bearer {{token}} (Inherit)

REQUEST BODY:

```
{
  "description": "Extra trip",
  "distance_km": 20,
  "emission_kg": 4.2
}
```

TEST SCRIPT:

```
var response = pm.response.json();
if (response.source) {
  pm.collectionVariables.set("new_source_id", response.source.source_id);
}
pm.test("Status code is 201", function () {
  pm.response.to.have.status(201);
});
pm.test("Source added successfully", function () {
  pm.expect(response.message).to.eql("Source added successfully");
});
pm.test("Source has source_id", function () {
  pm.expect(response.source).to.have.property("source_id");
});
```

3.2 Update Source

Method	URL	Auth
PUT	http://127.0.0.1:5001/activities/ACT-001/sources/{{new_source_id}}	Bearer {{token}} (Inherit)

REQUEST BODY:

```
{
  "description": "Updated morning commute",
  "distance_km": 70,
  "emission_kg": 14.7
}
```

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Source updated successfully", function () {
  var response = pm.response.json();
  pm.expect(response.message).to.eql("Source updated successfully");
});
```

3.3 Delete Source

Method	URL	Auth
DELETE	http://127.0.0.1:5001/activities/ACT-001/sources/{{new_source_id}}	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Source deleted successfully", function () {
  var response = pm.response.json();
});
```

```
pm.expect(response.message).toEqual("Source deleted successfully");  
});
```

4. Filtering

4.1 Filter by Type

Method	URL	Auth
GET	http://127.0.0.1:5001/activities/filter/type?activity_type=Transport	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Returns Transport activities only", function () {
  var response = pm.response.json();
  response.data.forEach(function(activity) {
    pm.expect(activity.activity_type).to.eql("Transport");
  });
});
```

4.2 Filter by Date Range

Method	URL	Auth
GET	http://127.0.0.1:5001/activities/filter/date?start_date=2026-01-01&end_date=2026-02-01	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Returns activities within date range", function () {
  var response = pm.response.json();
  pm.expect(response.data).to.be.an("array");
});
```

4.3 Filter by Minimum Emission

Method	URL	Auth
GET	http://127.0.0.1:5001/activities/filter/emission?min_emission=50	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("All activities above minimum emission", function () {
  var response = pm.response.json();
  response.data.forEach(function(activity) {
    pm.expect(activity.carbon_emission).to.be.at.least(50);
  });
});
```

4.4 Top Emitting Activities

Method	URL	Auth
GET	http://127.0.0.1:5001/activities/top-emitting	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Activities sorted by highest emission", function () {
```

```
var response = pm.response.json();  
for (var i = 0; i < response.data.length - 1; i++) {  
  pm.expect(response.data[i].carbon_emission).to.be  
    .at.least(response.data[i+1].carbon_emission);  
}  
});
```

5. Analytics

5.1 Total Emissions

Method	URL	Auth
GET	http://127.0.0.1:5001/analytics/total-emissions	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Has total emission data", function () {
  var response = pm.response.json();
  pm.expect(response.data[0]).to.have.property("total_emission_kg");
});
```

5.2 Emissions by Type

Method	URL	Auth
GET	http://127.0.0.1:5001/analytics/emissions-by-type	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Returns emission by type", function () {
  var response = pm.response.json();
  pm.expect(response.data).to.be.an("array");
  pm.expect(response.data[0]).to.have.property("total_emission_kg");
});
```

5.3 Monthly Trends

Method	URL	Auth
GET	http://127.0.0.1:5001/analytics/monthly-trends	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Returns monthly trend data", function () {
  var response = pm.response.json();
  pm.expect(response.data).to.be.an("array");
  pm.expect(response.data[0]._id).to.have.property("year");
});
```

5.4 Highest Activities

Method	URL	Auth
GET	http://127.0.0.1:5001/analytics/highest-activities	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Returns highest activity types", function () {
  var response = pm.response.json();
  pm.expect(response.data[0]).to.have.property("avg_emission_kg");
});
```

6. User Management

6.1 Get One User

Method	URL	Auth
GET	http://127.0.0.1:5001/users/USR-TEST	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("User data returned", function () {
  var response = pm.response.json();
  pm.expect(response.data).to.have.property("email");
});
pm.test("Password not exposed", function () {
  var response = pm.response.json();
  pm.expect(response.data).to.not.have.property("password");
});
```

6.2 Update User

Method	URL	Auth
PUT	http://127.0.0.1:5001/users/USR-TEST	Bearer {{token}} (Inherit)

REQUEST BODY:

```
{
  "name": "Test User Updated"
}
```

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("User updated successfully", function () {
  var response = pm.response.json();
  pm.expect(response.message).to.eql("User updated successfully");
});
```

6.3 Register Dummy User (for delete test)

Method	URL	Auth
POST	http://127.0.0.1:5001/users/register	No Auth

REQUEST BODY:

```
{
  "name": "Dummy User",
  "email": "dummy@email.com",
  "password": "password123",
  "role": "user"
}
```

TEST SCRIPT:

```
pm.test("Status code is 201", function () {
  pm.response.to.have.status(201);
});
pm.test("User registered successfully", function () {
  var response = pm.response.json();
  pm.expect(response.message).to.eql("User registered successfully");
});
```

6.4 Delete User

Method	URL	Auth
DELETE	http://127.0.0.1:5001/users/USR-DUM	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {  
  pm.response.to.have.status(200);  
});  
pm.test("User deleted successfully", function () {  
  var response = pm.response.json();  
  pm.expect(response.message).to.eql("User deleted successfully");  
});
```

7. RBAC Demonstration

7.1 Admin to See All Activities

Method	URL	Auth
GET	http://127.0.0.1:5001/activities	Bearer {{token}} (Inherit)

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Admin sees all activities", function () {
  var response = pm.response.json();
  pm.expect(response.total).to.be.above(0);
});
```

7.2 Register Normal User

Method	URL	Auth
POST	http://127.0.0.1:5001/users/register	No Auth

REQUEST BODY:

```
{
  "name": "Normal User",
  "email": "normal@email.com",
  "password": "password123",
  "role": "user"
}
```

TEST SCRIPT:

```
pm.test("Status code is 400 (email already exists)", function () {
  pm.response.to.have.status(400);
});
```

7.3 Login as Normal User

Method	URL	Auth
POST	http://127.0.0.1:5001/users/login	Basic Auth (normal@email.com)

TEST SCRIPT:

```
var response = pm.response.json();
pm.collectionVariables.set("normal_token", response.token);
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Normal user gets token", function () {
  pm.expect(response).to.have.property("token");
});
```

7.4 Get Activities with Normal User Token

Method	URL	Auth
GET	http://127.0.0.1:5001/activities	Bearer {{normal_token}}

TEST SCRIPT:

```
pm.test("Status code is 200", function () {
  pm.response.to.have.status(200);
});
pm.test("Normal user sees 0 activities", function () {
  var response = pm.response.json();
  pm.expect(response.total).to.eql(0);
});
```

```
});
```

55/55 tests passing | Basic Auth login | JWT for all other routes | RBAC confirmed

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